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Cosc 320-001

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Project 2 README

Approach:

Knowing the basic structure of a hashtable, I started creating it by making two structs (***Index***, and ***Node***) to represent the array of DLL that will make up the chain. A class called *dictionary* would hold an array of ***Index***, and the class functions would allow the program to perform necessary functions(e.g. search). This insight along with advice from a candy-throwing professor led me to get this project done relatively well.

Adding an extra character to a word-

I thought up a fairly neat way (in my opinion) to add a character to a word in each available position. It involves creating a new string with an extra `space` in the beginning (e.g. _poptart). Then I would circulate the alphabet in that empty space. Next I set the character back to a `space` and swapped it with the next character, where we iterate through our alphabet again. Repeat until done. Ex:

1. **_vex**-> **a**vex, **b**vex, **c**vex, **d**vex …
2. **v_ex**-> v**a**ex, v**b**ex, v**c**ex, v**d**ex …
3. **ve_x**-> ve**a**x, ve**b**x, ve**c**x, ve**d**x …
4. **vex_**-> vex**a**, vex**b**, vex**c**, vex**d** …

Code Issues/Limits:

- If you enter multiple misspelled words, and one cannot be corrected then you will go through a menu to correct some of the corrected words, but may not end up with a sentence correction.
- Limited mistake capacity of 20 misspelled words.
- Only one sentence will be checked.

Note: When compiling the program on two different machines (Linux & Windows), I received errors for `strdup`, so I compiled it with -gnu=c99 to get it to work.

By all means; it *should* work with:
g++ -std=c++11 -c main.cpp -g
g++ -o proj2.exe main.o -g
but, if not:
g++ -std=gnu11 -c main.cpp -g
g++ -o proj2.exe main.o -g